



SIMATS
ENGINEERING



SIMATS
Sree Vidya Institute of Medical And Technical Sciences
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SMART EMERGENCY PATIENT COORDINATION SYSTEM

**CSA0203 – C
Programming**

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Problem Statement



Hospital receives patients from multiple registration counters simultaneously.

Each patient has:

- Patient ID

- Emergency level

- Symptoms

- Assigned doctor

- Treatment status

Multiple departments may access the same patient simultaneously.

This can cause:

- Data conflicts

- Lost updates

- Incorrect patient information

- Record corruption



Objectives



Develop a reliable emergency patient coordination system.

Dynamically allocate memory for patient records.

Use functions for patient operations.

Use pointers to modify records.

Use threads to simulate hospital departments.

Use mutexes for safe concurrent updates.

Store patient history in files.

Use command-line arguments for department configuration.

Use IPC to transfer emergency information.



Proposed System



Centralized patient record management.

Multiple departments can work simultaneously.

Departments include:

- Registration

- Emergency/Triage

- Doctor/Treatment

- Pharmacy

- Discharge

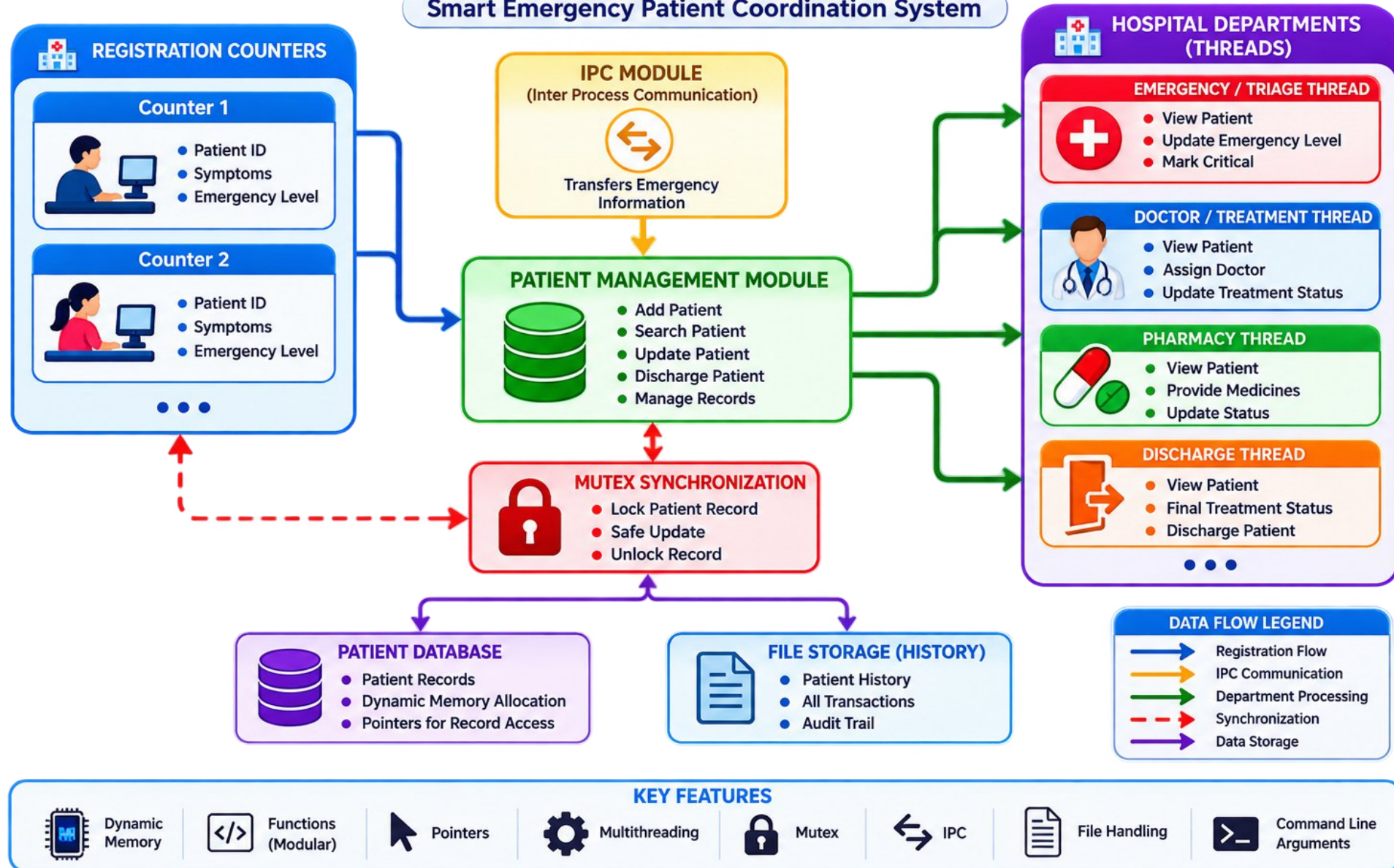
Real-time coordination between departments.

Patient records are protected during updates.

Complete history is maintained for every patient.

SYSTEM ARCHITECTURE

Smart Emergency Patient Coordination System





Patient Data Management



C Structures – Store complete patient records.

Patient Information – ID, emergency level, symptoms, doctor, and treatment status.

Dynamic Memory Allocation – Allocates memory as patients are added.

Core Functions – Register, Search, Update, and Discharge patients.

Pointers – Used to efficiently access and modify patient records.

Benefit – Flexible and efficient patient data management.



Multithreading



POSIX threads are used to simulate hospital departments.
Each department operates as a separate thread.

Example:

- Registration Thread

- Emergency Thread

- Doctor Thread

- Pharmacy Thread

- Discharge Thread

Multiple departments can process different patients simultaneously.

Threads improve system responsiveness.

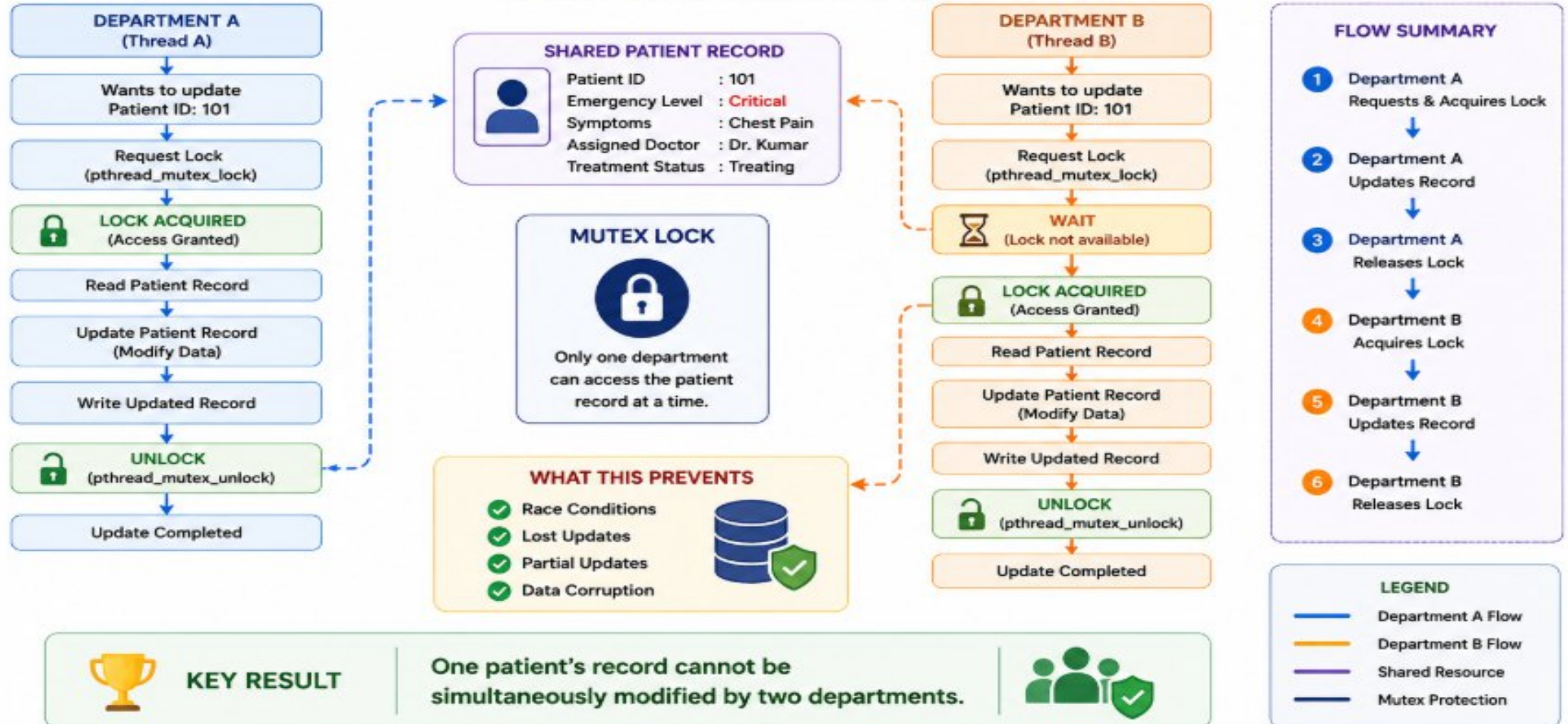
Shared patient data requires synchronization.

Challenge



CHALLENGE: Prevent two departments from overwriting the same patient record simultaneously.

MUTEX WORKFLOW FOR UPDATE





IPC & File Handling



IPC connects the registration and treatment modules.

Registration sends:

- Patient ID

- Emergency level

- Symptoms

Treatment module receives the emergency information.

Patient history is stored in a file.

History includes:

- Registration

- Emergency-level changes

- Doctor assignment

- Treatment updates

- Discharge

Provides an audit trail of patient activities.



Future Scope



AI-based emergency-level prediction.

IoT-based patient monitoring.

Mobile application.

SMS/emergency notifications.

Cloud-based hospital integration.

Ambulance tracking.

Advanced analytics.

Electronic Health Record integration.

```
Registered Patient 1 with Doctor Dr. Smith  
IPC: Sent patient 1 to treatment module  
Department 3 active...  
Department 2 active...  
Department 1 active...  
Updated Patient 1 status to Under Treatment  
Updated Patient 1 status to Under Treatment  
Updated Patient 1 status to Under Treatment  
Discharged Patient 1
```



Conclusion



Smart system for coordinating emergency patient information.

Supports multiple hospital departments.

Uses dynamic memory, pointers, functions and threads.

Mutex protects shared patient records.

IPC enables module-to-module communication.

File handling maintains patient history.

Successfully addresses the hackathon challenge.